Lecture 8: Panel Data & Time Series

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Applied Analytical Statistics

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Addendum | Two reasons to add controls in regression

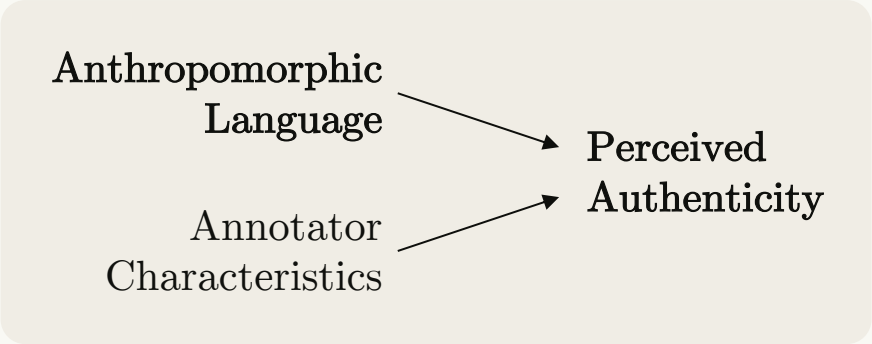
Controls for **identification**: block backdoor paths, remove OVB, enable causal interpretation

Controls for **precision**: reduce **unexplained variance** and thus lower SE of treatment effect.

$\sigma^2 = \mathrm{Var}(\varepsilon)$ which influences $\mathrm{Var}(\beta)$

treatment

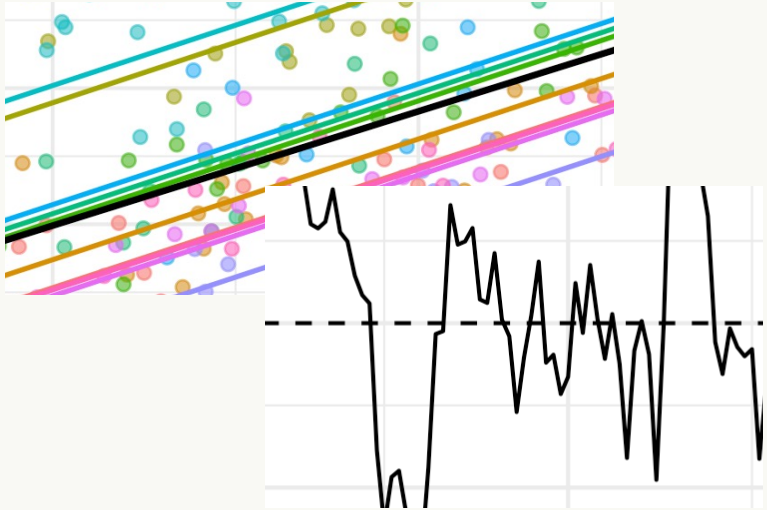
not a confounder BUT an important predictor of the outcome



outcome

By adding relevant controls, we avoid our treatment being buried in noisy outcomes.
What remains is "cleaner" variation attributable to our treatment.

Plan for today | Panel data and time series



Today, we focus on regression models for **structured data**: panels and time series.

We need to account for this structure to do **valid inference**.

We can leverage this structure to **address endogeneity concerns**.

We will finish the lecture by going through **summative structure and expectations**.

Panel data | Structure

Panel data is data that tracks the same $N > 1$ units over $T > 1$ time periods.

worker AI use and productivity across months

subreddit toxicity metrics before/after policy change

misinfo engagement across platforms

social media use across countries and years

There are two dimensions of variation:

Between variation = variation across units

Within variation = variation over time for the same unit.

cross-sectional data: $T = 1$

time series data: $N = 1$

A **balanced** panel includes observations for all units for all time periods.
An **unbalanced** panel is missing observations for some units or time periods.

Panel data | Unobserved heterogeneity

We can exploit panel structure to address the problem of **unobserved heterogeneity**.

RQ: Is using AI associated with being more productive?

e.g. worker skill

Units differ from each other in ways that are **relevant to both regressor and outcome**.
→ **omitted variable bias** undermines statistical inference and identification.

In **cross-sectional data**, we observe each unit once.
We can only control for variables we observe.

In **panel data**, we observe each unit multiple times.
We can control for unobserved heterogeneity!

Panel data | Fixed effects

Fixed effects (FE) regression controls for all **time-invariant unobserved heterogeneity**.

$Y_{it} = \beta X_{it} + \alpha_i + u_{it}$ where α_i is the unobserved unit-specific effect.

By including unit fixed effecs α_i, we capture all **time-invariant between-unit variation**.
This includes observed and unobserved characteristics!

Data: Worker-level AI use (weekly hours) and productivity across 20 weeks

Assume: skill predicts AI use and productivity (OVB) <u>and</u> we do not observe it ☹
By including worker FE, we can still control for it!

Fixed effects | Implementation with dummy variables

We can estimate FE linear regression by including $N - 1$ unit dummies and using OLS:

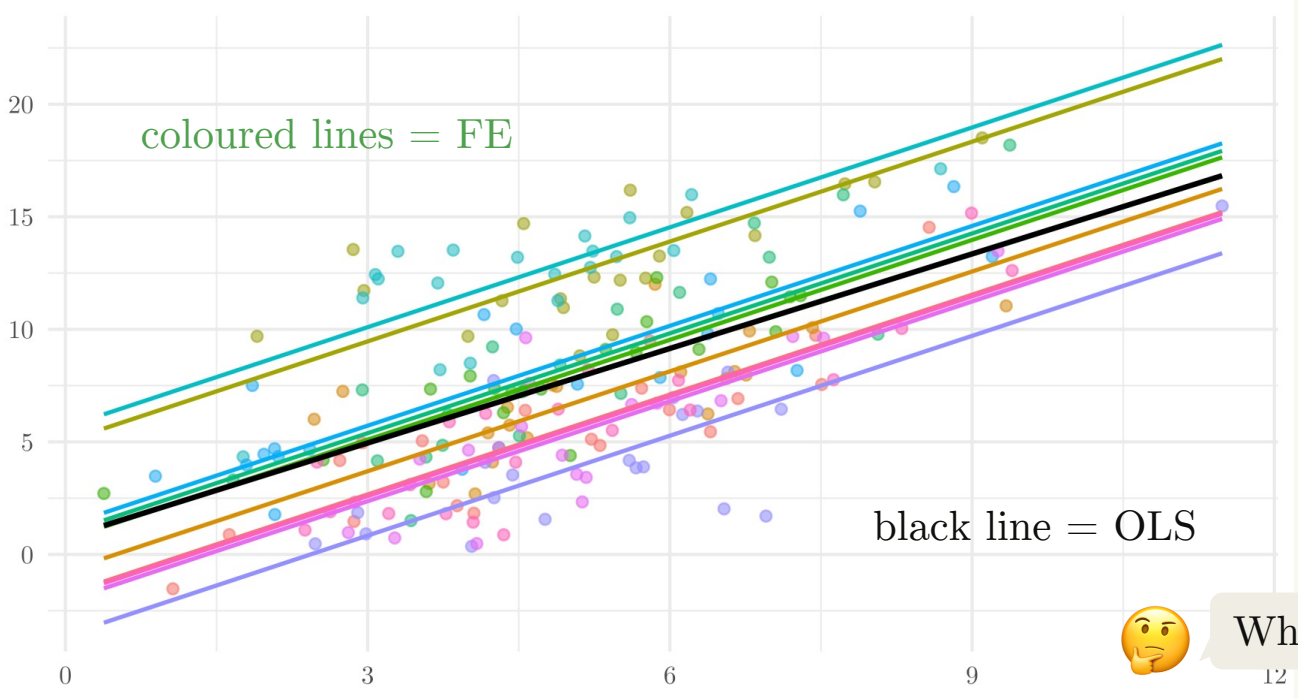
$$Y_{it} = \beta X_{it} + \sum_i \gamma_i D_i + u_{it}$$

where $D_i = 1$ if observation belongs to unit i

🤔 Why not include N dummies?

We omit one dummy as a **reference category** to avoid perfect multicollinearity.
All other dummies are interpreted relative to the reference category.

Fixed effects | Graphical intuition



Including α_i gives **each unit i** its own intercept.

The slope β is the same across all units.

🤔 Why is $\beta_{FE} \neq \beta_{OLS}$?

Fixed effects | Within-unit variation

By including unit FE α_i, we capture all time-invariant between-unit variation.
This means **we estimate $\boldsymbol{\beta}$ only based on within-unit variation!**

(Week 5)

Remember **Frisch-Waugh-Lovell**: the OLS coefficient on X is estimated using only the variation in X that is **orthogonal** to the other regressors, which now include FE dummies.

i.e. not linearly explained by

Estimating FE with dummy variables is equivalent to fitting:

$$Y_{it} - \overline{Y}_i = \beta(X_{it} - \overline{X}_i) + (u_{it} - \bar{u}_i)$$

the "Within Transformation"

FE asks: "When a unit's X moves away from its usual level, does Y move with it?"

$\rightarrow$ each unit is compared only to itself

Fixed effects | Shift in interpretation

Without FEs, we ignore panel structure. OLS coefficients are interpreted as before.

$Y_{it} = \beta X_{it} + \varepsilon_{it}$ also known as "**pooled OLS**", uses within- and between-unit variation.

e.g. "Do workers who use more AI tend to be more productive?"

With **unit FEs**, we only use within-unit variation:

$Y_{it} = \beta X_{it} + \alpha_i + u_{it}$ also known as the "**within estimator**".

e.g. "When a worker increases AI use, is this associated with an increase in their productivity?"

(Collapsing the panel to unit averages gives us "**between estimator**" $\overline{Y}_i = \beta \overline{X}_i + v_i$)

e.g. "Do workers who, on average, use more AI tend to, on average, be more productive?"

Fixed effects | Time-invariant covariates

Time-invariant regressors cannot be estimated in a unit FE model.

$Y_{it} = \beta X_{it} + \alpha_i + u_{it}$ where α_i is the unobserved unit-specific effect.

This is because time-invariant regressors are **perfectly collinear** with the unit FE α_i.

$\rightarrow$ time-invariant regressor is a linear combination of the unit dummies

Data: Worker-level AI use (weekly hours) and productivity across 20 weeks

Regressors: weekly AI use, race, education level, experience

Weekly AI use and experience can vary ✓
Education level and race are fixed ✗

🤔 Which of these can we estimate?

Fixed effects | Limited within-unit variation

When there is little within-unit variation, we have little information to estimate β.

Data: Worker-level AI use (weekly hours) and productivity across <u>2</u> weeks.

Many workers may not use AI in either week or use AI equally much in both weeks.

For T=2, we are effectively estimating on one observation per worker:

$\Delta Y_i = \beta \Delta X_i + \Delta u_i$ so that $\Delta Y_i = \Delta u_i$ when $\Delta X_i = 0$

$\rightarrow$ Units i without variation in X are completely irrelevant to estimating β.

This is why you should **beware of short panels** (small T):
Within-unit means are noisy and possible within-unit variation is limited.

Fixed effects | Losing degrees of freedom

Standard errors generally depend on degrees of freedom:

$\hat{\sigma}^2 = \text{RSS}/\text{df}$ in linear regression

$\rightarrow$ width of confidence intervals

$\rightarrow$ power of hypothesis tests

Without FEs, degrees of freedom are determined by breadth and length of our panel:

$\text{df} = \text{NT} - (\text{K} + 1)$ where $\text{K} =$ number of regressors

When we include unit FEs using dummy variables, we reduce df by $\text{N} - 1$.

This is a second reason why you should **beware of short panels**:
We lose a larger proportion of df from including unit FEs when T is small.

Fixed effects | Time-varying confounding

Unit fixed effects remove all **time-invariant** heterogeneity between units.
However, **heterogeneity may change over time!**

RQ: Is using AI associated with being more productive?

If an **omitted variable changes over time**, and this variable is **relevant**, then OVB persists!

e.g. worker experience grows

i.e. correlated with X_{it}

$E(u_{it} \mid X_{i1}, \dots, X_{iT}, \alpha_i) \neq 0$: the strict exogeneity assumption is violated.

For **long panels** (large T) time-invariant heterogeneity is a strong assumption.
Few things are "fixed" over long time periods.

Fixed effects | Time fixed effects

We can apply the same FE logic to the **time dimension** in our panel.

$Y_{it} = \beta X_{it} + \lambda_t + u_{it}$ where λ_t is the unobserved time-specific effect.

Time FEs control for **unit-invariant temporal heterogeneity**: "shocks" that affect <u>all</u> units.

Data: Worker-level AI use (weekly hours) and productivity across 20 weeks

Assume: GPT-6 is released in week 10.

🤔 Do we need time FE to avoid OVB?

YES if the release has a direct effect on AI use and productivity → backdoor path!

NO if the release affects productivity only via increased AI use.

Fixed effects | Two-way fixed effects

Finally, we can combine **unit FEs** and **time FEs**:

$Y_{it} = \beta X_{it} + \alpha_i + \lambda_t + u_{it}$ controls for both time- and unit-invariant heterogeneity

By including **unit FEs**, we remove all time-invariant unit differences.
By including **time FEs**, we remove all time-specific common shocks.

β is now estimated based on **within-unit variation over time, net of common time shocks**.
We compare changes in each unit to its own average and the average change in that period.

Data: Worker-level AI use (weekly hours) and productivity across 20 weeks

e.g. "When a worker increases their AI use in a given week relative to their average level, and relative to other workers in that same week, does their productivity increase?

Fixed effects | Beyond panel data

We can apply FE to many kinds of data with natural "groupings", not just panels!

RQ: Does chain-of-thought prompting lead to better LLM responses?

5 LLMs × 2 prompting strategies × 100 benchmark questions.

Reasonable assumptions: LLMs differ in overall quality. Questions differ in difficulty.

$Y_{ij} = \beta X_{ij} + \alpha_i + \lambda_j + u_{ij}$ controls for model-specific and question-specific heterogeneity.

where Y_{ij} is model i's score on question j.

regions platforms countries

Many of your summative datasets will have at least one natural grouping! annotators

Consider FE, but beware of consequences for estimation, interpretation, ... years

Brief aside | Random effects

We can also use random effects (REs) instead of unit-level fixed effects:

$Y_{it} = \beta X_{it} + \alpha_i + u_{it}$ where $\alpha_i \sim N(0, \sigma_\alpha^2)$: unit-level RE is drawn from normal dist.

This introduces within-unit correlation without estimating a per-unit parameter.

RE is much **more efficient than FE**: i.e. smaller standard errors

- We can use both within- and between-unit variation for estimation.
- We add one variance parameter σ_α^2 instead of N intercepts.

With REs, we can **include time-invariant regressors** in our model.

However, RE is unbiased and consistent only under an even stronger **exogeneity assumption**:

$Cov(X_{it}, \alpha_i) = 0$: RE is uncorrelated with regressors X

cannot solve OVB for
UsesAI ← Skill → Productivity

Fixed effects | Clustered standard errors

The **independence assumption** (of i.i.d.):

$\mathrm{Cov}(u_{it}, u_{js}) = 0$ for $(i,t) \neq (j,s)$ is rarely plausible in panel data.

Within units, **serial correlation** is common: unobserved traits evolve, shocks persist over time.

$\mathrm{Cov}(u_{it}, u_{js}) \neq 0$ for $t \neq s$

→ correlated errors are less informative

Violation of the independence assumption leads to **incorrect SEs and invalid inference**.

We fix this problem using **clustered standard errors** that correct for within-unit correlation.

Clustered SEs treat each unit as an independent block of information: N clusters.
Degrees of freedom are closer to N than NT, which can create problems for small N.

Time series | Structure

Time series are a special case of panel data where $N = 1$ and $T > 1$.

single firm productivity across months

toxicity of single subreddit before/after policy change

number of misinfo posts on a single platform

UK social media use across years

There is no between-unit variation. All dependence is temporal.

In time series, we typically assume that **independence is violated**:

$\mathrm{Cov}(u_t, u_{t-k}) \neq 0$ i.e. errors correlate with lagged errors = **autocorrelation**

This leads to **incorrect standard errors** and **invalid inference**.

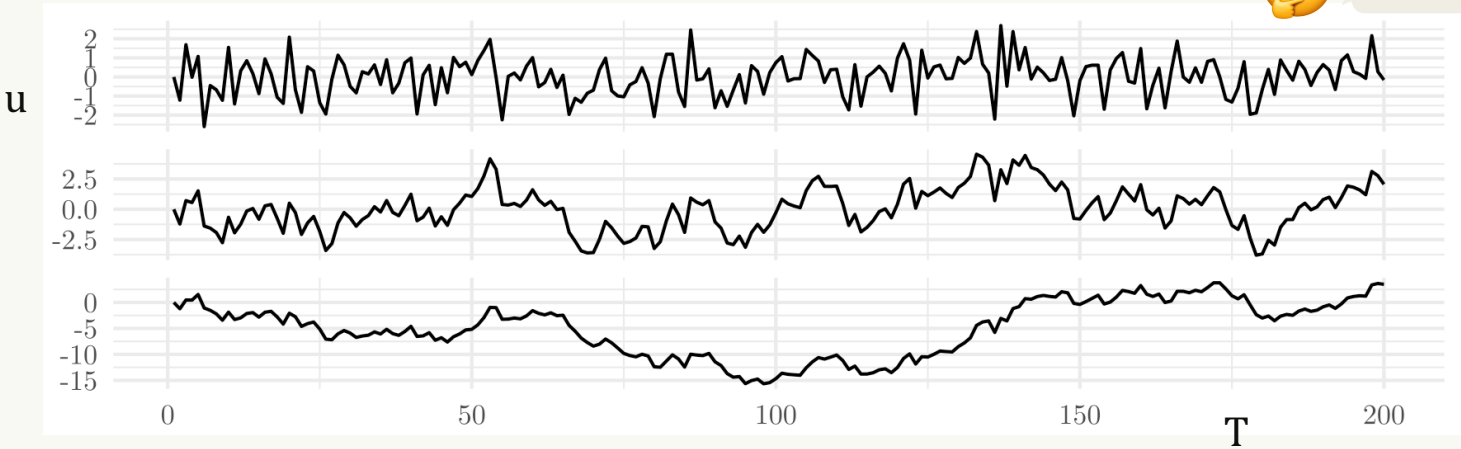
Time series | The AR(1) model

The simplest mathematical model of temporal dependence is an **AR(1) model**:

$u_t = \rho u_{t-1} + \varepsilon_t$ i.e. an **A**uto**R**egressive model that includes **1** lagged value of **u**.

where ρ measures **persistence over time**.

Can you guess ρ?



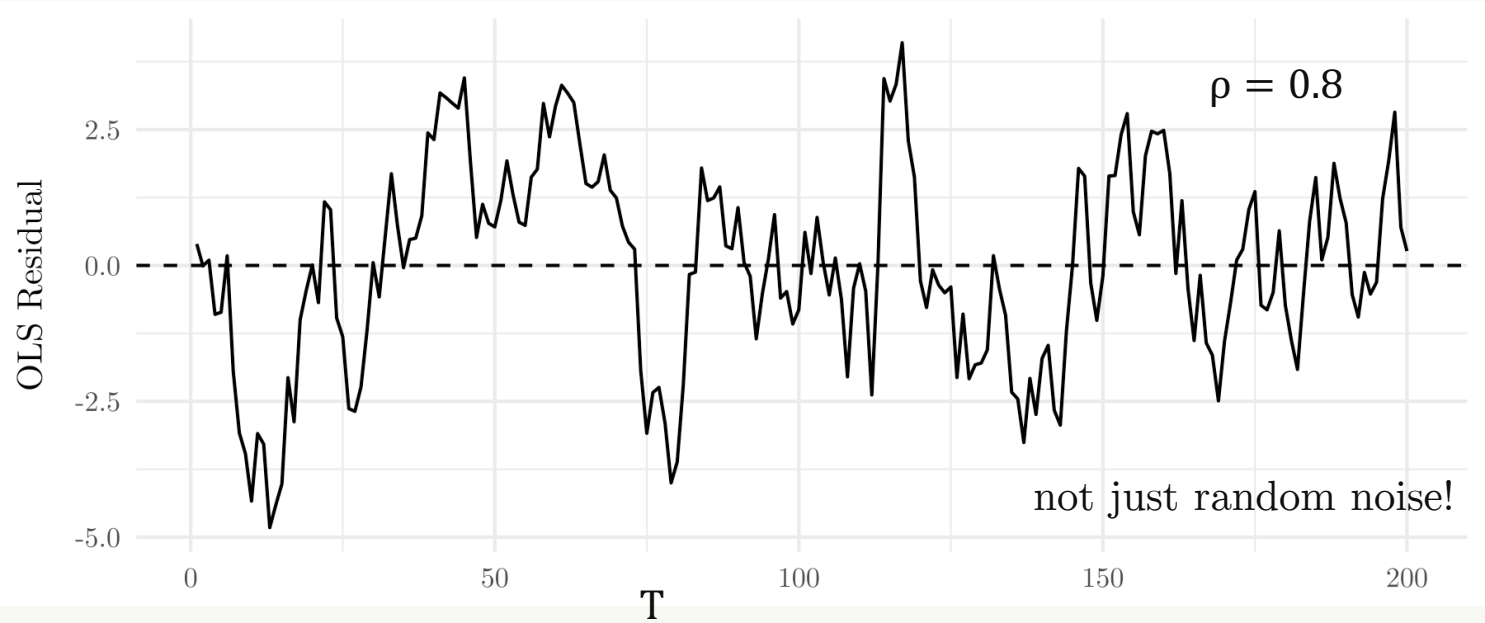
$\rho = 0$
white noise

$\rho = 0.8$

$\rho = 1$
random walk

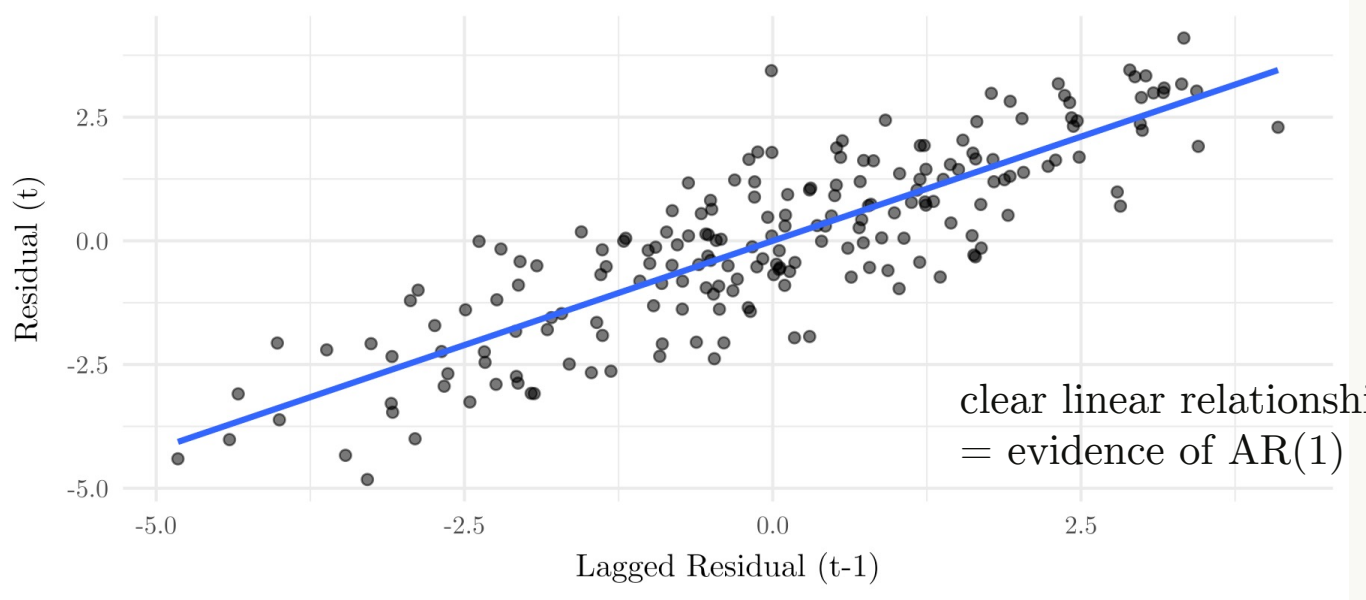
Time series | Diagnosing autocorrelation

After fitting OLS, we can spot autocorrelation as **persistence in residuals $\hat{u}_t$ over time**.



Time series | Diagnosing autocorrelation (cont'd)

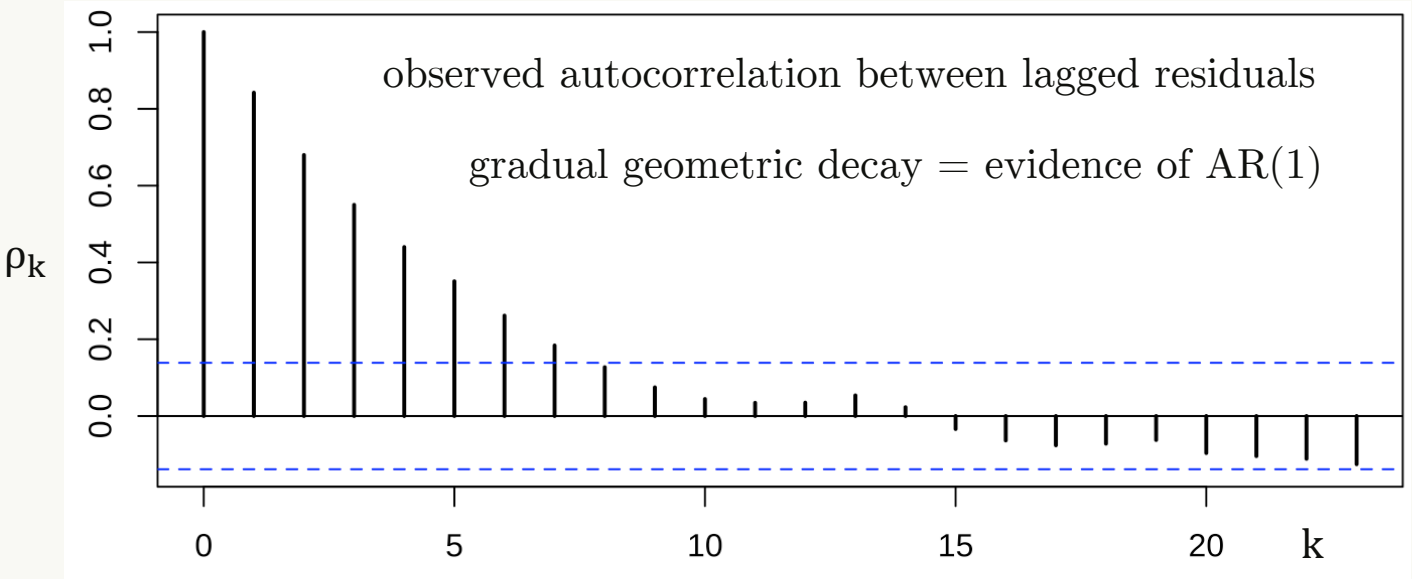
If we suspect an AR(1) process, we can **compare subsequent residuals**:



clear linear relationship
= evidence of AR(1)

Time series | Diagnosing autocorrelation (cont'd)

Finally, the **autocorrelation function (ACF)** for errors $\rho_k = \mathrm{Corr}(u_t, u_{t-k})$



Time series | Durbin-Watson test

The **Durbin-Watson test** provides a formal test for first-order autocorrelation.

$$\mathrm{DW} = \frac{\sum_{t=2}^{T} (\hat{u}_t - \hat{u}_{t-1})^2}{\sum_{t=1}^{T} \hat{u}_t^2}$$

which, for large T $\approx 2(1 - \hat{\rho})$

The test statistic DW takes values between 0 and 4.
It does not have a well-behaved null distribution, so we rely on rules of thumb:

$\mathrm{DW} \approx 2 \rightarrow$ no first-order autocorrelation

$\mathrm{DW} < 2 \rightarrow$ positive first-order autocorrelation
$\mathrm{DW} > 2 \rightarrow$ negative first-order autocorrelation

🤔 Can you guess what $\mathrm{DW} < 2$ means?

Time series | HAC standard errors

Suppose $Y_t = \beta X_t + u_t$ where $\mathrm{Cov}(u_t, u_{t-k}) \neq 0$ and $E(u_t | X_1, \dots, X_T) = 0$

Errors are persistent, but strict exogeneity holds so that there is no OVB.
OLS is unbiased, but **SEs are incorrect** and **inference is invalid**.

→ correlated errors are less informative

We can fix this using **H**eteroskedasticity and **A**utocorrelation **C**onsistent standard errors.

HAC standard errors use squared residuals (for heteroskedasticity) and products of residuals across time lags (for autocorrelation) to produce a "sandwich" estimator:

$\mathrm{Var}(\hat{\beta}) = (X'X)^{-1} X'\Omega X (X'X)^{-1}$ where $X'\Omega X$ is estimated by HAC

Time series | Modelling persistence

If dynamics are substantively meaningful, we may want to model them directly:

$Y_t = \beta X_t + \phi Y_{t-1} + u_t$ where ϕ measures **persistence in outcomes**.

Now β measures short-run effects of X_t, conditional on past outcomes Y_{t-1}.

If persistence is very strong ($\phi \approx 1$), we can estimate **first differences** instead:

$\Delta Y_t = \beta \Delta X_t + \varepsilon_t$ where $\Delta Y_t = Y_t - Y_{t-1}$, which removes persistence.

Here, β measures change in Y associated with change in X over subsequent time periods.

In both cases, we can use OLS or MLE to estimate model coefficients.

Recap | Key takeaways from Week 8

Panel data contains two kinds of variation.
Variation between units at the same time and variation within units across time.

Fixed effects control for observed and unobserved heterogeneity.
Unit FEs remove time-invariant heterogeneity and time FEs remove common shocks.

Structured data violates the independence assumption.
Serial correlation in panels and time series leads to incorrect SEs if not accounted for.

Dependence and temporal dynamics can be modelled explicitly.
We can adjust SEs or model dynamics directly in AR models with lagged outcomes.

Summative | Structure

Students must submit a **4,000-word research paper**. The paper should include the following components:

- An **introduction** that motivates the project, states the main research question, and situates the project within relevant academic literature.
- A **data** section describing the dataset used, how it was collected, and why it is suitable for answering the research question.
- A **methods** section detailing the statistical techniques used for analysis.
- A **results** section reporting the findings, using visualisations where appropriate.
- A **discussion** that draws conclusions, discusses their implications in the context of relevant literature, and reflects on limitations of the analysis.

Summative | Research question and motivation

Section: **Introduction**

This is where you **define what your project is about**.

Clearly state the **research question(s)** you are answering.
The question should be precise and answerable based on your analysis.

Explain **why it is interesting / useful** to answer this question.
What is the relevance of your findings to society or industry or policy or ... ?

Relate your work to **prior work** that answers similar questions.
What research gap are you filling? What evidence are you extending?

Summative | Data description

Section: **Data**

This is where you communicate the **shape of your data** and **how it came into existence**.

Clearly **state key facts** about your data.
What is the unit of observation? What is the sample size? What are the variable types?

Use **descriptive stats** and (where appropriate) **visualisations.**
How is your outcome variable distributed? How are the regressors distributed?

Describe **how data was collected** and how this relates to **data quality**.
What was the sampling process? When was data collected? How were variables measured?

Summative | Statistical design and model choice

Section: **Methods**

Clearly state **which models/methods** you will use to answer the RQ.
What are the regression equations? Which are the coefficients of interest?

Explain the **reasoning behind your statistical design**.
Why is this model/method appropriate, given the data you are working with?
Why are you including or excluding specific variables?

State the **assumptions behind your statistical design**.
Under which conditions does your model/method produce valid results?

Preference: (variants of) models covered in this course rather than complex specialised models.

Summative | Statistical reasoning and inference

Section: **Results**

This where you **describe, i.e. state, the results** of your main analyses.
- Give clear interpretations for coefficients of interest. This will depend on your design!
- Comment on uncertainty of estimates: CIs, p-values, statistical significance.
- Compare size and significance of coefficients across different model specifications.

Make sure to **draw only appropriate statistical conclusions** based on your design.
- Do not use causal language if unwarranted. Remember sample vs. population distinction.

Summative | Diagnostics and robustness

Section: **Results** (usually after main results)

Assess whether the **assumptions** behind your statistical design are plausible.
Functional form, error distributions, independence... This will depend on your design!

Explore whether your results are **sensitive to reasonable modelling choices**.
Alternative model specifications, different sets of controls, different operationalisations...

If possible, describe the **overall fit** of your model.
Does your model explain the data well? Does this change across specifications?

You can move detail to the Appendix if you reference it clearly from the main body.

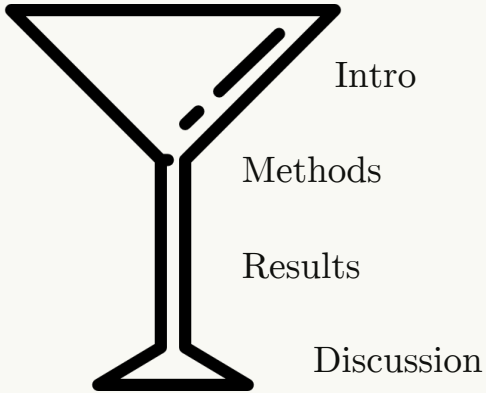
Summative | Interpretation and limitations

Section: **Discussion**

This is where you connect back your **specific findings** to the **broader narrative**.

Interpret the substance of your findings.
  Are your results (e.g. coefficients) practically significant?
  What are the overall conclusions about your RQ?

Discuss the **limitations** of your analysis and resulting claims.
  What are potential sources of error, e.g. in your data?
  Are there alternative explanations for your findings?
  To what extent would you expect findings to generalise?



Summative | Clarity

Sections: **all of them!**

You write for the reader, not for yourself. Paper writing is communication.

Make the **structure** of your work visible to the reader.
Give important ideas their own paragraphs. Use informative sub-headings.

Use **clear and direct language**. Dense ≠ rigorous. Formal ≠ precise. Long ≠ sophisticated.
Use active voice. If a sentence must be read twice, rewrite it!

Use **visualisations** strategically. Each Table or Figure must answer a clear question.
Avoid redundancy. Use interpretable colours, readable font sizes, etc.

Summative | Code

Submitted as /**code folder** in submission.zip

Please use R for all statistical analysis since this is the focus of this course.

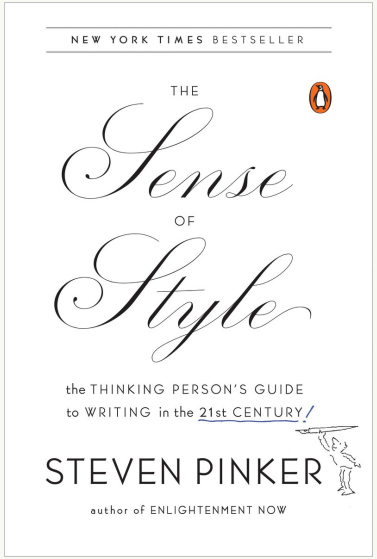
Your code must be **well-documented**, i.e. easy to understand by others.
Think of the code folder as a GitHub repo: include readme.md, clear naming, etc.

Your code must be **reproducible** with minimal effort.

Do not include large data files in your submission. Link to GitHub / Drive / ... instead.

Well-documented, reproducible code is a **requirement**, not a positive marking criterion.

Beyond the summative | Advice on good writing



"The starting point for becoming a **good writer** is to be a **good reader**. Writers acquire their technique by spotting, savouring, and reverse-engineering examples of good prose."

"The guiding metaphor of classic style is **seeing the world**. The writer can see something that the reader has not yet noticed, and he orients the reader's gaze so that she can see it for herself"

"Good writing takes advantage of a **reader's expectations** of where to go next. It accompanies the reader on a journey, or arranges the material in a logical sequence."

Beyond the summative | Advice on writing research papers

https://nicholas.carlini.com/writing/2026/how-to-win-a-best-paper-award.html

"The single most important skill to develop for high-impact research is **good taste in what problems are worth solving**. If you have good taste, and you keep writing papers, eventually you'll write one with exceptionally high impact."

"**Pick your ideas for impact**. [...] You need to go into research with the objective of discovering something interesting, important, and new. The paper follows as a natural consequence."

"**Do something only you can do** [and] find your **comparative advantage**"

Above all: "**get lucky**" in finding good ideas, at the right time, in the right space, ...

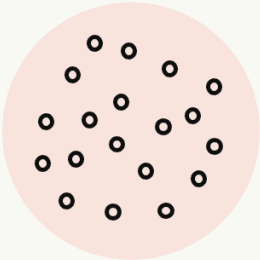
Course recap | Key concepts and ideas

populations vs samples

statistics as random variables

LLN & CLT

sampling distributions & standard errors



regression as model of conditional mean

exogeneity & OVB

control variables & FWL logic

functional form

generalised linear models

i.i.d

data-generating process

identification & exchangeability